

Video Transcript

Video 7: MongoDB: A Document Database (09:24)

At a very high level, a data store is a data store. That is, we store information within it. However, when it comes to the conceptual model that you have to think of as a user of that database, as a developer that is going to be storing information within the database, a relational database management system is very different than a document database. Let's explore a little bit more what we mean by that, especially when it comes to getting started within a database design, a document database is quite different. It is simply a collection of documents, and those documents can have anything you like within them.

Now, that document collection would be the equivalent of a table in a relational database. However, it doesn't have the structure or the schema. And once we get to the code, that will become more apparent. Then you would have within this document, your fields and your properties and this would be the equivalent of columns on a relational database. And ultimately that third part, which is the relationships, is how the collections and the properties interact. Now, there are some great advantages when it comes to document data stores, and that is that the documents are really independent units.

Again, as mentioned previously, you don't have that structure, that hard schema that really limits a lot of what you're doing, especially at the beginning of a project when you are exploring. Then when it comes to the application logic, because of it, as you get going, you don't have to conform to that structure. Now, it might be that you need it and that the functionality of a relational database is really something that you need from the very get go, but this is really within the design space. It depends on what you're trying to do. If you are simply exploring and you're looking to work with a data store, that's really not going to get in your way while you figure things out, document database is perfect.

And as you scale up, it might be that, that continues to be the case. However, there are some trade-offs. So you can see here that one of the biggest advantages is not having that structure pushed upon you. Now, when it comes to thinking about a document, what do we really mean by a document? When you look at this diagram here, you can see that what we really mean is really just a JSON document. You can see here that is a collection of value pairs and what you have

within it is really up to you, meaning that you have the flexibility of having different documents that have different properties and different fields within them, even if they're all within the same collection.

Now when it comes to being able to create these, as you might imagine, being able to do so means getting a handle on the database on the table, which in this case is the collection, and then inserting into that. The same thing would be true when it comes to reading back. You get a handle on the database, the collection, and then you pull out what you need. Now, in this case, when it comes to the document database, the one that we're going to be using is MongoDB. This is a widely used, very popular open source database, and once again, we're going to be putting it into a container and accessing it locally on our machine.

The command that we're going to be used to create that Docker container inside of on top of the platform docker is the following. You can see there that we have Docker run at the very end. It's the image name, which is Mongo, and between it we have the port, the name that we're going to give it, and that we're going to run this container, detached. So as you can see here, I have a terminal window opened up at my desktop, and I'm going to go ahead and run that command and I'm going to go ahead and open up the dashboard to be able to see when that is created.

I'm going to go ahead and make this a little bit smaller so you can see, and I'm going to clear my desktop to get rid of that output that you saw there. And now I'm going to go ahead and run the command. And as you can see there, I am doing a fresh install that was not found locally. I am pulling that container and in a moment, as that gets instantiated and pulled into my local environment, you can see here at the top that I now have a Mongo database up and running and I can go ahead and connect to it.

Now the next thing that we're going to do is that we're going to write some Python code against this new container, this new Mongo database. But before we do so, we need to install the driver. And so I'm going to do precisely that. I'm going to go ahead and clear here the desktop and I'm going to go ahead and install that driver. As you can see there that had already been installed. So I simply get the confirmation that I have already done so. So now let's go ahead and write some code.

I'm going to go ahead and open up my editor. I'm going to go ahead and drag and drop a sample folder that I have on my desktop. And once that opens, you can see there that I have two files. One is called read and the other one is called write. We'll start off with write, and once we do

write into the database, then we're going to go ahead and use the second file to be able to read that back. Once we open the file, we're going to go ahead and start off by adding or importing the packages we need. You can see that the very first one there is the driver.

The second one is what we're going to use to create a timestamp that we're going to write into the database as well as one for the unique identifier. Right after that, we're going to go ahead and make a connection and note here that all we're doing is simply writing the name of the database. Then we specify that is in the localhost and the port that you see there is the default port. Right after that, we're going to get a handle on the collection. This is or not the collection, but on the database. And in this case we're going to call it Pluto.

It's worth noting that if that does not exist, it's going to be created. This is a fresh, clean installation of the database, and this is one of those advantages that I mentioned. It creates simply what it does not have. So as opposed to what you see in relational where you need to create things first, create schemas, create tables, and so on, here it'll simply create it for you. Now the next thing after that is going to be the collection of posts under Pluto under that database. And once again, it'll do the same thing. This is the equivalent of the table and it'll simply create the table.

Now next comes where we're going to define a JSON document, and as you can see there, I am creating an ID with the unique ID UUID. And then we're going to go ahead and create a timestamp. And I am creating a JSON structure there, which is really a dictionary within Python. And then the last thing that we're going to do is that we're going to insert that into the table again, the collection of documents called posts. So let's go ahead and save this document.

Let's go ahead and open up the console. And as you can see there, I am on the desktop. I'm going to go ahead and enter here the name of the file, which is right. And as you can see there, that executed without any errors. So now let's go ahead and write that read file. We'll start off once again by getting a handle on the driver. We'll then make our connection to Mongo. We'll get a handle on the database, then the collections, then we're going to read through all of them. And that will be all that we will be doing.

So let's go ahead and run that program. This one is called Read, and as you can see there, I'm going to go ahead and expand the window. We can see that date stamp that we wrote with that unique identifier. Now, note that Mongo itself adds on its own a unique identifier of its own. And that is what that other item that you see there. Now let's go ahead and make a few more 'Write'.

I'm going to go ahead and run two more, and then let's go ahead and read it again. And as you can see there now we have 3 entries.

We have those 3 timestamps, note the time differences there. We have 3 unique IDs that we created, plus the ones that Mongo itself is creating. So now it's your turn. Make sure you can create the document database container. Make sure you can connect to it, make sure you can write, make sure you can read.

